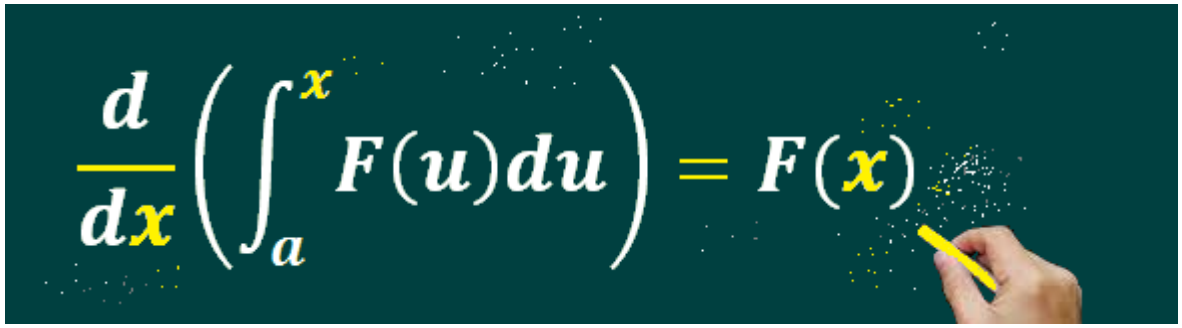
Refresher Course in Mathematics 2023

Instructor:



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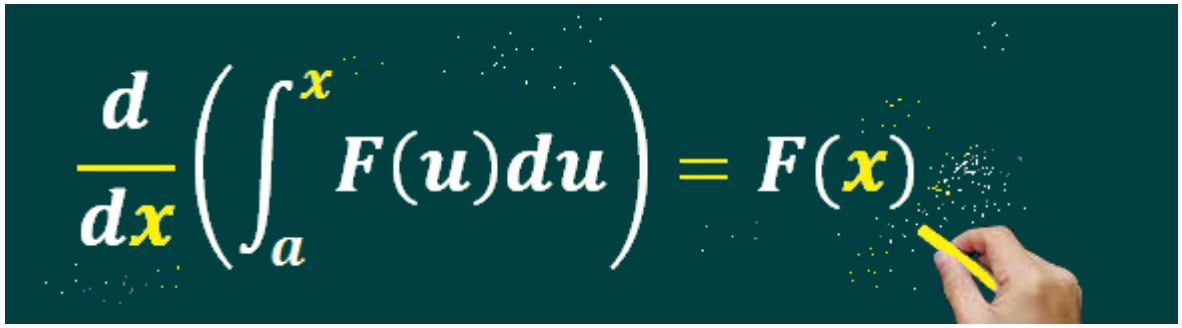
We will go through topics in elementary functions and single variable calculus.

The level of the course is perched somewhere between your pre-university and first year undergraduate mathematics.

Download the PDF entitled “Lessons in Elementary Functions and Single Variable Calculus” from the course website at NTULearn.

A password is needed to access the document.
The password is**: isaacnewton**





Session One

Lessons in Elementary Functions and Single Variable Calculus
Chapter 1, Sections 1.1 and 1.2



Real numbers

positive integers (whole numbers)

1, 2, 3, 4, 5, ... for counting

Arithmetic operations + − × ÷

$5+3=8$ $5\times 2 = 2+2+2+2+2 = 10$

$5-4=1$

Because of substractions like 4-5, 8-10, 1-4, ..., we have negative integers

-1, -2, -3, ...

Substractions like 1-1, 2-2, 3-3, ... gives the integer 0 (zero).



Real numbers

Integers $1, 2, 3, 4, 5, \ldots$

$-1, -2, -3, -4, -5, \ldots$

0

Divisions such as $1 \div 2$, $3 \div 9$, $(-1) \div 4$, $(-3) \div (-4)$, ..., give rise to rational numbers $1/2$, $3/9$, $-1/4$, $3/4$, ...

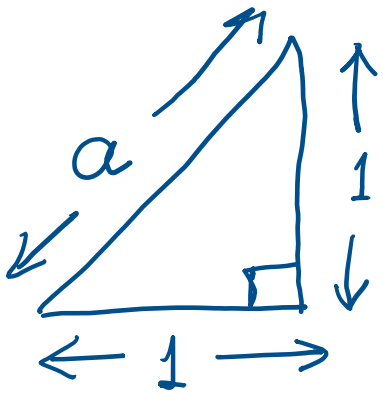
A rational number can be eventually written in the form m/n, where m and n ($n \neq 0$) are integers that do not have any common integer factors except -1 and 1.

$3 = \frac{-3}{-1} = \frac{3}{1}$

E.g. $\frac{3}{9} = \frac{1}{3}$ $\frac{3}{(-4)} = \frac{-3}{4}$ $-3 = \frac{-3}{1}$



Real numbers



According to Pythagoras theorem

$$a^2 = 1^2 + 1^2 = 2$$

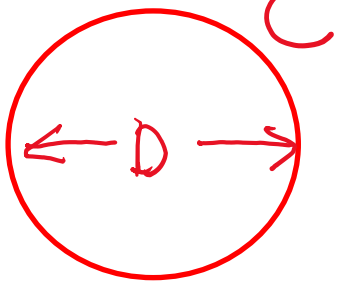
Let us denote a by $\sqrt{2}$.

So $\sqrt{2}$ is a number such that

$$\sqrt{2} \cdot \sqrt{2} = 2.$$

$\sqrt{2}$ cannot be written in the form p/q, where p and q are integers.

$\sqrt{2}$ is not a rational number.



Circle

$\frac{C}{D} = \pi$ π is an irrational number

$$e = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots\ldots$$

e is irrational

$-\sqrt{2}$

$-\pi$ $-e$



Real numbers

integers $1, 2, 3, 4, \ldots$
$-1, -2, -3, -4, \ldots$
0

Rational numbers $-\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{4}{3}, \ldots, m/n \quad (n \neq 0)$

Integers are special rational numbers

Irrational numbers

E.g. $\pi, -\pi, \sqrt{2}, -\sqrt{2}, e, -e, \ldots$



Real numbers in decimal form

In decimal form, a real number comprises the sign of the number, a non-negative integer before a decimal point, and an infinite number of non-negative integers after the decimal point.

$$\frac{1}{3} = 0.3333333\cdots = 0.\overline{3}$$

$$\frac{5}{4} = 1.2500000\cdots = 1.25\overline{0}$$

$$\frac{22}{15} = 1.4666666\cdots = 1.4\overline{6}$$

$$\frac{41}{333} = 0.123\,123\,123\cdots = 0.\overline{123}$$

$$\frac{431}{135} = 3.1\,925\,925\,925\,925\,925\cdots = 3.1\,\overline{925}$$

Decimal digits

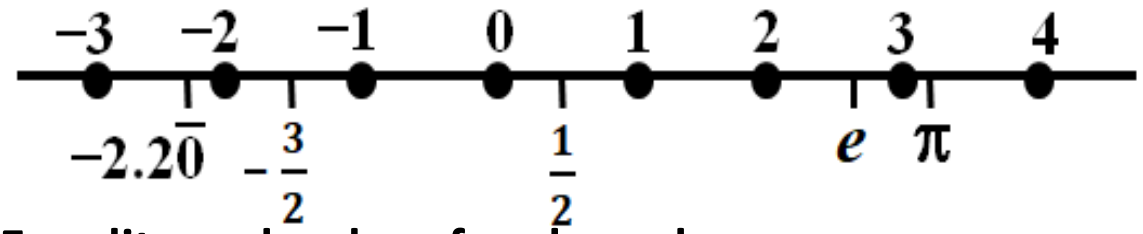
Decimal digits of a rational number have a predictable pattern (after a certain decimal digit).

$\pi = 3.14159265358979323846264338 \ldots\ldots$

Decimal digits of an irrational number do not have a predictable pattern.



Real number line



Real number line is DENSE.

Equality and order of real numbers

Let a and b be two real numbers.

We say that $a = b$ if a and b occupies the same position on the real number line.

$a < b$ if a is on the left of b on the real no. line

$a > b$ if a is on the right of b on the real no. line.

$1 < 2$ $\quad -2 < -1$

$2 > 1$ $\quad -1 > -2$

$x > 1$ $\quad x \geq 1$

$x < \pi$ $\quad x \leq \pi$



Absolute value of a real number

Let a be a real number.

We define:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

absolute value of a

$|2| = 2$

$|-3| = -(-3) = 3$

$|a|$ is a non-negative real number

$|a|$ = distance of a from 0 on the real no. line

$|b-a| = |a-b|$

$|5-3| = |3-5| = 2$

$|b-a| = |a-b|$ = distance between a and b on the real no. line



Intervals

An interval is the set of real numbers lying on a portion of the real no. line.

$\{x: a < x < b, x \text{ is real}\} = (a, b)$



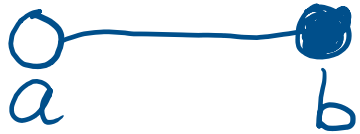
$(1, 2) = \{x: 1 < x < 2, x \text{ is real}\}$

$\{x: a \le x \le b, x \text{ is real}\} = [a, b]$



$[-1, 1] = \{x: |x| \le 1, x \text{ is real}\}$

$\{x: a < x \le b, x \text{ is real}\} = (a, b]$



$(0, 3] = \{x: 0 < x \le 3, x \text{ is real}\}$

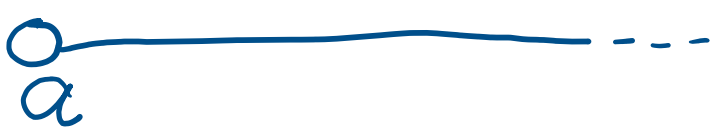
$\{x: a \le x < b, x \text{ is real}\} = [a, b)$

$[0, 3) = \{x: 0 \le x < 3, x \text{ is real}\}$



Intervals

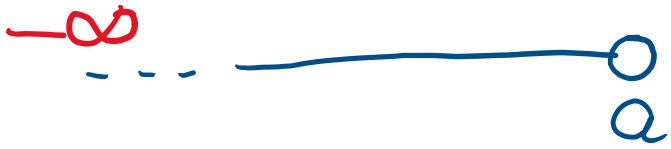
$\{x : x > a, x \text{ is real}\} = (a, \infty)$



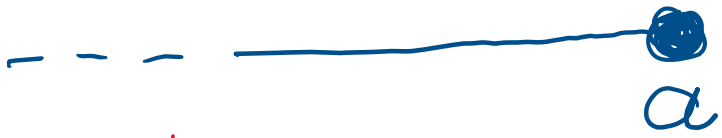
$\{x : x \geq a, x \text{ is real}\} = [a, \infty)$



$\{x : x < a, x \text{ is real}\} = (-\infty, a)$



$\{x : x \leq a, x \text{ is real}\} = (-\infty, a]$



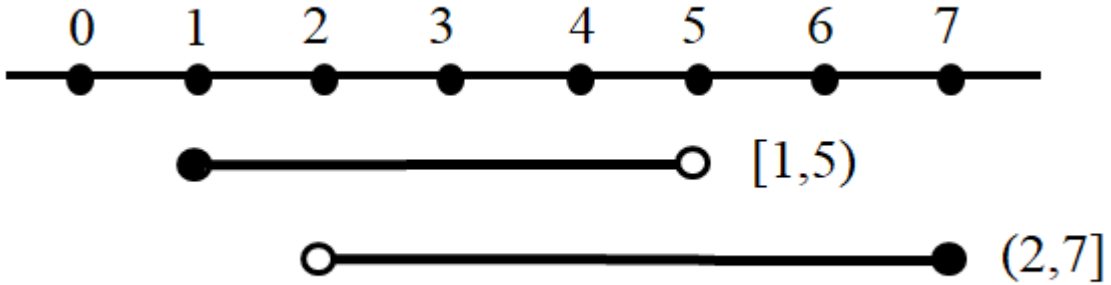
$(-\infty, \infty)$ = the set of all real numbers



Intervals

Intervals are sets. So we can use set notations for them.

$\frac{1}{2} \in (0,1)$ $\frac{1}{2} \notin [\frac{1}{4}, \frac{1}{2})$



$[1,5) \cap (2,7] = (2,5)$

$[1,5) \cup (2,7] = [1,7]$



Real sequences

A well-ordered set of real numbers given by $\{a_1, a_2, a_3, \ldots, a_N\}$ is called a real sequence

The real numbers $a_1, a_2, \ldots, a_N$ are called the elements of the sequence.

The order in which the elements appear in the sequence is important.

The sequences $\{1, 7, 5, 3\}$ and $\{3, 1, 7, 5\}$ are different sequences although they contain the same set of real numbers in them.



Real sequences

A real sequence is said to be infinite if it has infinitely many elements.

E.g. $\{0, 2, 4, 6, 8, \ldots\} = \{a_k\}$ where $a_k = 2k-2$ for $k = 1, 2, 3, \ldots$

An infinite sequence $\{a_1, a_2, \ldots\}$ is said to have a limit L, where L is a fixed real number, if $|a_k - L|$ becomes closer and closer to zero as k increases.

If $\{a_k\}$ has limit L, we write $\lim_{k\to\infty} a_k = L$.

$\{0, 2, 4, 6, 8, \ldots\}$ does not have a limit since $a_k = 2k-1$ does not get closer and closer to any fixed real no. as k increases.

$\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$

$= \{a_k\}$ where $a_k = \frac{1}{k}$ for $k = 1, 2, 3, \ldots$ has limit 0 since $\lim_{k\to\infty} \frac{1}{k} = 0$



Real sequences

Consider the infinite real sequence $\{a_1, a_2, a_3, \ldots\}$ where a_k is the rational number such that the integral part of a_k (in decimal form) is 3, the first k decimal digits of a_k are the same as those of π, and the remaining decimal digits are all zero.

$\pi = 3.14159265358979323846264338327 \ldots\ldots$

$$a_1 = 3.1\overline{00} = \frac{31}{10}$$

$$a_2 = 3.14\overline{00} = \frac{157}{50}$$

$$a_3 = 3.141\overline{00} = \frac{3141}{1000}$$

$$a_4 = 3.1415\overline{00} = \frac{6283}{2000}$$

$$\vdots$$

As k increases, a_k gets closer and closer to π.

$\lim_{k\to\infty} a_k = \pi$

$\Rightarrow \{a_k\}$ has limit given by π.



Series

Consider an infinite real sequence $\{a_1, a_2, a_3, \ldots\}$.

We use the elements of $\{a_k\}$ to form the series

$$a_1 + a_2 + a_3 + \cdots = \sum_{k=1}^{\infty} a_k$$



We define the m-th partial sum of the series $a_1 + a_2 + a_3 + \cdots$ by

$$S_m = a_1 + a_2 + a_3 + \cdots + a_m = \sum_{k=1}^{m} a_k$$

$S_1 = a_1$, $S_2 = a_1 + a_2$, $S_3 = a_1 + a_2 + a_3$, $\ldots$

We have the sequence $\{S_1, S_2, S_3, \ldots\}$

If $\{S_k\}$ has limit L, then the series $a_1 + a_2 + a_3 + \cdots$ is said to converge and we write $a_1 + a_2 + a_3 + \cdots = L$.

Convergent series

Otherwise, the series is said to diverge.

Divergent series

Convergent and divergent series

Consider the sequence $\{1, -1, 1, -1, \ldots\} = \{a_k\}$

where $a_k = \begin{cases} 1 & \text{if } k \text{ is odd} \\ -1 & \text{if } k \text{ is even} \end{cases}$

We form the series $1 - 1 + 1 - 1 + 1 - 1 + \ldots$

The m-th partial sums of this series are:

$S_1 = 1, \quad S_2 = 1 - 1 = 0, \quad S_3 = 1 - 1 + 1 = 1, \quad S_4 = 1 - 1 + 1 - 1 = 0,$

$\ldots \Rightarrow S_k = 1$ if k is odd

$S_k = 0$ if k is even

The sequence $\{S_k\}$ does not have a limit.

$\Rightarrow$ The series $1 - 1 + 1 - 1 + 1 - 1 + \ldots$ diverges.

The terms in a series have to be summed up one by one at a time according to their order of appearance.

$1 - 1 + 1 - 1 + 1 - 1 + \ldots \neq (1 - 1) + (1 - 1) + (1 - 1) + \ldots$



Convergent and divergent series

If $\lim_{k\to\infty} a_k \neq 0$ then the series

$a_1 + a_2 + a_3 + \cdots$ diverges.

If $\lim_{k\to\infty} a_k = 0$ then the series $a_1 + a_2 + a_3 + \cdots$ may either converge or diverge depending on what a_k is.

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{4} \quad (\approx 1.644934)$$

$$S_{1000} = \sum_{k=1}^{1000} \frac{1}{k^2} \approx 1.643934$$

$$S_{2000} = \sum_{k=1}^{2000} \frac{1}{k^2} \approx 1.644434$$



Convergent and divergent series

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

diverges even though

$$\lim_{k \to \infty} \frac{1}{k} = 0$$

$$S_{1000} = \sum_{k=1}^{1000} \frac{1}{k} \approx 7.485$$

$$S_{2000} = \sum_{k=1}^{2000} \frac{1}{k} \approx 8.178$$

$$S_{4000} \approx 8.871 \qquad S_{8000} \approx 9.564$$

$$S_{16000} \approx 10.257$$



Exponents or powers



We will now look at the definitions of a^p. We refer to p as the exponent or power of a^p.

Positive integer exponents

Let a be any real number and n be a positive integer.

We define a^n as follows.

$a^1 = a$

$a^n = a \times a^{n-1}$ for $n = 2, 3, 4, \ldots$

$a^2 = a \times a^1 = a \times a$

$a^3 = a \times a^2 = a \times a \times a$

In general,

$a^n = \underbrace{a \times a \times \ldots \times a}_{n \text{ times}}$

If p and q are positive integers then $a^p \times a^q = a^{p+q}$.

$a^2 \times a^3 = a \times a \times a \times a \times a = a^5$

Exponents or powers



Negative integer exponents

Let a be a non-zero real number and n be a positive integer.

We define $a^{-n} = \frac{1}{a^n}$ $(a \neq 0)$.

E.g. $2^{-4} = \frac{1}{2^4} = \frac{1}{2\times2\times2\times2} = \frac{1}{16}$

$(-5)^{-3} = \frac{1}{(-5)^3} = \frac{1}{(-5)\times(-5)\times(-5)} = -\frac{1}{125}$

If p and q are positive integers

then $\frac{a^p}{a^q} = a^{p-q}$ $(a \neq 0)$ $(p \neq q)$

$\frac{a^2}{a^5} = \frac{a\times a}{a\times a\times a\times a\times a} = \frac{1}{a^3} = a^{-3}$

We have not defined a^0 yet!

Zero exponent

$\frac{a^p}{a^q} = a^{p-q}$ Suppose we let $p = q$ in this property.

$$a^{p-q} = a^0 = \frac{a^p}{a^p} = 1 \quad (a \neq 0)$$

So we define:

$$a^0 = 1 \quad \text{for } a \neq 0$$

So far, we have defined:

$a^1, a^2, a^3, \ldots$ for all real no. a

$a^{-1}, a^{-2}, a^{-3}, \ldots$ } for real no. $a \neq 0$

a^0



Roots

Exponents of the form $1/n$, where n is an integer such that $n \geq 2$

We will now define $a^{1/n}$ for integers $n \geq 2$.

Start with $n = 2$, i.e. $a^{1/2}$

square root of a

$\sqrt{a}$

Let $a^{1/2} = y$.

POSITIVE ← **real number**

Then we want y to be the real number such that $y^2 = a$.

We know y^2 cannot be negative for real y.

$\Rightarrow a^{1/2}$ can only be defined for $a \geq 0$.

E.g. Suppose we want to define $4^{1/2}$.

Let $4^{1/2} = y$.

$y^2 = 4$

We can have either $y = 2$ or $y = -2$

We choose $4^{1/2} = +2$



Roots



Let a be a non-negative real number.
We define $a^{1/2}$ to be the positive real y such that $y^2 = a$.

$a^{1/2} \cdot a^{1/2} = a$ for $a \geq 0$.

Let us now look at $a^{1/3}$.

cube root of a: $\sqrt[3]{a}$

We want $a^{1/3}$ to be the real number y such that $y^3 = a$.

If $y \leq 0$, $a \leq 0$. If $y \geq 0$, $a \geq 0$.

No restriction on a in the definition of $a^{1/3}$.

Also, the sign of y is automatically determined by the sign of a.

Roots



E.g. $8^{1/3} = y$ y is such that $y^3 = 8$

We can only have $y = 2$.

$$8^{1/3} = 2$$

$(-8)^{1/3} = y$ y is such that $y^3 = -8$.

$(-8)^{1/3} = -2$

We can have only $y = -2$

Let a be any real number.

We define $a^{1/3}$ to be the real number y such that $y^3 = a$.

$a^{1/3} \cdot a^{1/3} \cdot a^{1/3} = a$

Roots $a^{1/n}$ for $n = 2, 3, 4, \ldots$ n-th root of a $\sqrt[n]{a}$

Let a be a positive real number and n be a positive even integer ($n = 2, 4, 6, 8, \ldots$). We define $a^{1/n}$ to be the positive real number y such that $y^n = a$.

Let a be any real number and n be a positive odd integer greater than 2 ($n = 3, 5, 7, 9, \ldots$). We define $a^{1/n}$ to be the real number y such that $y^n = a$.



Rational exponents

Let m be any integer and n be a positive integer.

It can be shown that

$$(a^m)^{1/n} = (a^{1/n})^m$$

We may have to impose a restriction on a

E.g.

$(a^{1/2})^3 = x \Rightarrow a^{1/2} = x^{1/3} \Rightarrow a = (x^{1/3})^2$

$(a^3)^{1/2} = y \Rightarrow a^3 = y^2 \Rightarrow ((x^{1/3})^2)^3 = y^2$

$\Rightarrow (x^{1/3})^3)^2 = y^2$

$\Rightarrow x^2 = y^2$

$\Rightarrow x = y$

$a \geq 0$

$x \geq 0$

$y \geq 0$

Define:

$$a^{m/n} = (a^{1/n})^m = (a^m)^{1/n}$$



Rational exponents

E.g. $8^{2/3} = (8^2)^{1/3} = 64^{1/3} = 4$

$8^{2/3} = (8^{1/3})^2 = (2)^2 = 4$

$(-32)^{2/5} = ((-32)^{1/5})^2 = (-2)^2 = 4$

$(-32)^{2/5} = ((-32)^2)^{1/5} = (1024)^{1/5} = 4$

$$a^{-m/n} = \frac{1}{a^{m/n}} \quad (a \neq 0)$$

So far, we have defined a^r for all rational exponents r.



Irrational exponents



Recall that an irrational number is a real number that cannot be written in the form m/n, where m and n are integers.

How do we define a^ρ if the exponent ρ is an irrational number?

Let $\{r_1, r_2, r_3, \ldots\}$ be an infinite real sequence of rational numbers such that r_k has limit given by the irrational number ρ as k increases, i.e. $\lim_{k\to\infty} r_k = \rho$.

Since r_k is a rational number, a^{r_k} is defined.

So the sequence $\{a^{r_1}, a^{r_2}, a^{r_3}, \ldots\}$ can be formed.

We define a^ρ as the limit of the sequence $\{a^{r_1}, a^{r_2}, a^{r_3}, \ldots\}$, i.e. $a^\rho = \lim_{k\to\infty} a^{r_k}$.

E.g.

Consider the infinite real sequence $\{a_1, a_2, a_3, \ldots\}$ where a_k is the rational number such that the integral part of a_k (in decimal form) is 3, the first k decimal digits of a_k are the same as those of π, and the remaining decimal digits are all zero.

$\pi = 3.14159265358979323846264338 \ldots\ldots$

$$a_1 = 3.1\overline{00} = \frac{31}{10}$$

$$a_2 = 3.14\overline{00} = \frac{157}{50}$$

$$a_3 = 3.141\overline{00} = \frac{3141}{1000}$$

$$a_4 = 3.1415\overline{00} = \frac{6283}{2000}$$

$$\vdots$$

As k increases, a_k gets closer and closer to π.

$\lim_{k\to\infty} a_k = \pi$

$\Rightarrow \{a_k\}$ has limit given by π.

$2^{\pi} = \lim_{k\to\infty} 2^{a_k}$



Proofs are given in Section 1.2

Properties (laws) of exponents

Let p, q, u, v be any real numbers.

#1. If u^p and u^q exist then

$$u^p \cdot u^q = u^{p+q}.$$

#2. If $(u^p)^q$ exists then

$$(u^p)^q = u^{pq}.$$

#3. If u^p and v^p exist then

$$u^p \cdot v^p = (uv)^p$$

$$\frac{u^p}{v^p} = \left(\frac{u}{v}\right)^p.$$



Any question?



A yawning long-tailed macaque at Singapore Chinese Gardens
Image courtesy of Robert Ang of “I walked, I spotted, I captured”